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Predicting Exchange Rate Using LSTM Models and Multi-Source News Sentiment

Introduction

The foreign exchange market is highly volatile, influenced by various factors such as economics, politics, and market sentiment, making short-term exchange rate movements a significant challenge. Among all currency pairs, the EUR/USD is one of the most actively traded, its fluctuations directly reflecting the economic rivalry between the Eurozone and the United States. Because of its importance, a reliable method to predict its direction the following day would be invaluable to traders and quantitative systems.

Our project aims to explore combining historical price data with sentiment analysis from financial news to see if we can more accurately predict the EUR/USD's movement the following day.

Traditional methods like ARIMA often focus solely on price curves, neglecting textual information like news reports. However, the market reacts very quickly to news, policies, and even media tone. Many traders check Yahoo Finance, ForexLive, and the ECB's website daily because these platforms not only report facts but also convey sentiment and attitudes that

influence expectations. Incorporating these sentiment signals into the model should improve predictive performance, especially in the short term.

So our core idea is simple: combine news sentiment with LSTM models, which excel at handling time series data, to see if we can improve prediction accuracy. Specifically, we scrape articles from several major news sources for the day, score each article using the VADER sentiment analysis tool, and then aggregate them into a daily sentiment index. Finally, we feed this sentiment index along with historical price data into an LSTM model to predict the exchange rate for the next trading day. For price data, we use Yahoo Finance. In short, we want to demonstrate that deep learning models that consider sentiment can indeed predict more accurately than models that only look at prices.

Our entire approach can be divided into several steps: data collection, text processing, sentiment analysis, feature integration, and finally, neural network modeling for prediction. Below, I'll explain each step in detail, focusing on where and how VADER sentiment analysis is implemented.

Methodology

Data Preparation

The first step is finding news. We primarily scrape financial news related to the EUR/USD exchange rate from three sources: Yahoo Finance's RSS subscription, ForexLive's real-time news, and the European Central Bank's official press releases. These three sources were chosen because they each have their own focus: Yahoo Finance represents mainstream viewpoints, ForexLive provides intraday news updates for forex traders, and the ECB's press releases offer the most authoritative policy information. We collected news from the week of

November 18th to 25th, 2025, ensuring the timing of the news matched the exchange rate data we used.

After obtaining the news, it cannot be used directly; it needs to be "washed," meaning the text is preprocessed. We wrote a script to convert all text to lowercase, remove punctuation, and then use the NLTK toolkit to remove stop words like "the" and "a" from English text. Finally, we restored words to their original forms (e.g., "running" becomes "run"). This was done to allow the sentiment analyzer to better grasp the core meaning of the article, rather than being distracted by irrelevant noise.

Text processing

The next crucial step is scoring the news using VADER. In the code, we directly import `SentimentIntensityAnalyzer` from the `vaderSentiment` library and create an analyzer instance. Whenever we process a cleaned-up news article, we call a function that passes the text to the analyzer's `polarity_scores()` method and takes only the "compound score." This score, ranging from -1 to +1, accurately represents whether the article is generally positive, negative, or neutral. VADER is a tool naturally suited for analyzing short texts like social media posts and has been validated in the financial sector, making it perfect for scoring news headlines and short reports. The score and publication date of each article are stored together. The average score of all articles published on the same day is then calculated to obtain the day's sentiment index. This generates three sentiment curves corresponding to the three news sources.

Sentiment analysis

Sentiment data alone is insufficient; price data must also be included. Historical EUR/USD prices are obtained from Yahoo Finance, coded as "USDEUR=X". Note that this represents the USD to EUR exchange rate, so we need to divide 1 by its closing price to calculate

the EUR/USD exchange rate. After calculation, MinMaxScaler is used to normalize the price data to the range of 0-1, facilitating neural network training. Finally, the price and sentiment data are aligned by date, resulting in a large table with four columns: processed EUR/USD price, Yahoo's sentiment index, ForexLive's sentiment index, and the ECB's sentiment index.

Feature integration & Neural network modeling

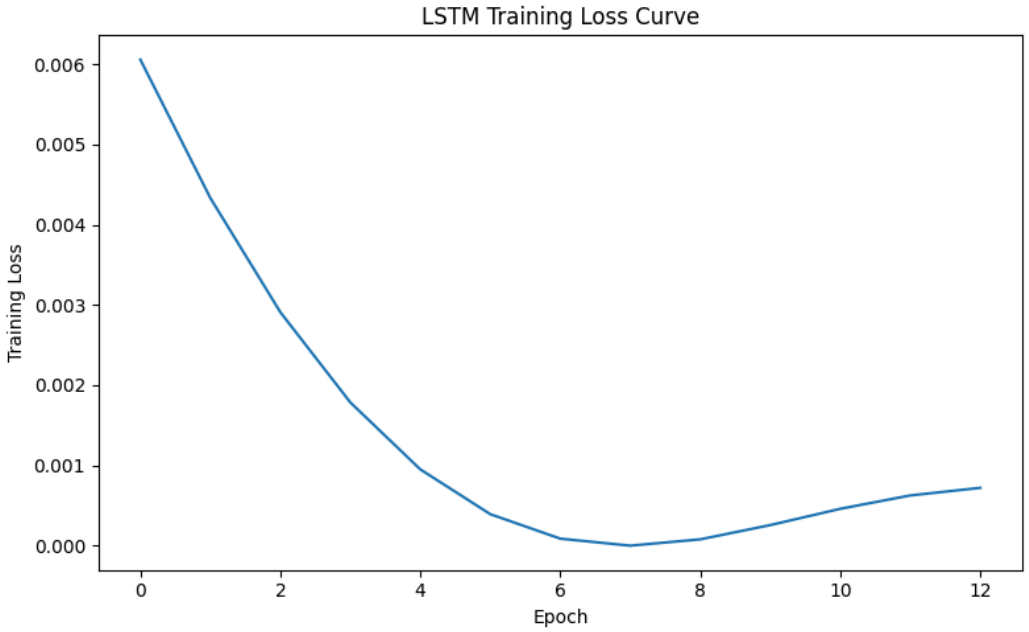
Finally, modeling and prediction are performed. We use an LSTM neural network because it excels at processing time-series data and can remember previous information. Our model is quite simple: a single LSTM layer with 50 neurons, followed by a fully connected layer, and finally outputting a single value—the predicted exchange rate for the next day. We let the model use data from the past 1 to 7 days to make predictions. To prevent the model from becoming "overfitted," we used early stopping; if the performance didn't improve after 5 consecutive training iterations, it would automatically stop. The optimizer used was Adam, with a learning rate of 0.001, and the loss function was mean squared error (MSE). After the model was trained, we input the latest data window to obtain the predicted value for the next day. The complete code for all these steps is in the Python script we submitted.

Results

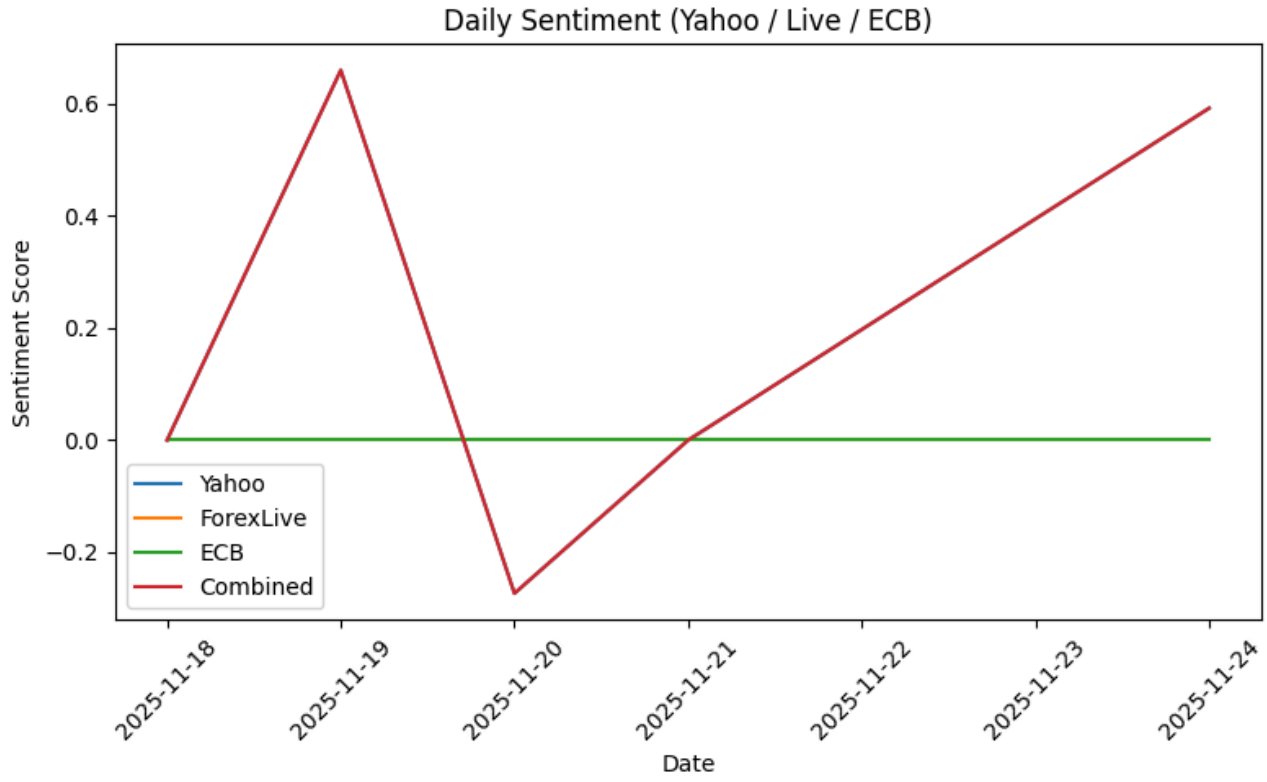
The results of this study include visualizations of the historical price curve, the aggregated daily sentiment curves from all three news sources, and the LSTM model’s training loss progression. These results help illustrate both the data characteristics and the performance of the forecasting model.

The first figure depicts the LSTM training loss curve, showing how the network’s loss decreases over the training epochs. The loss decreases rapidly in the early epochs and continues to decline until reaching a plateau near zero, indicating stable convergence and successful early

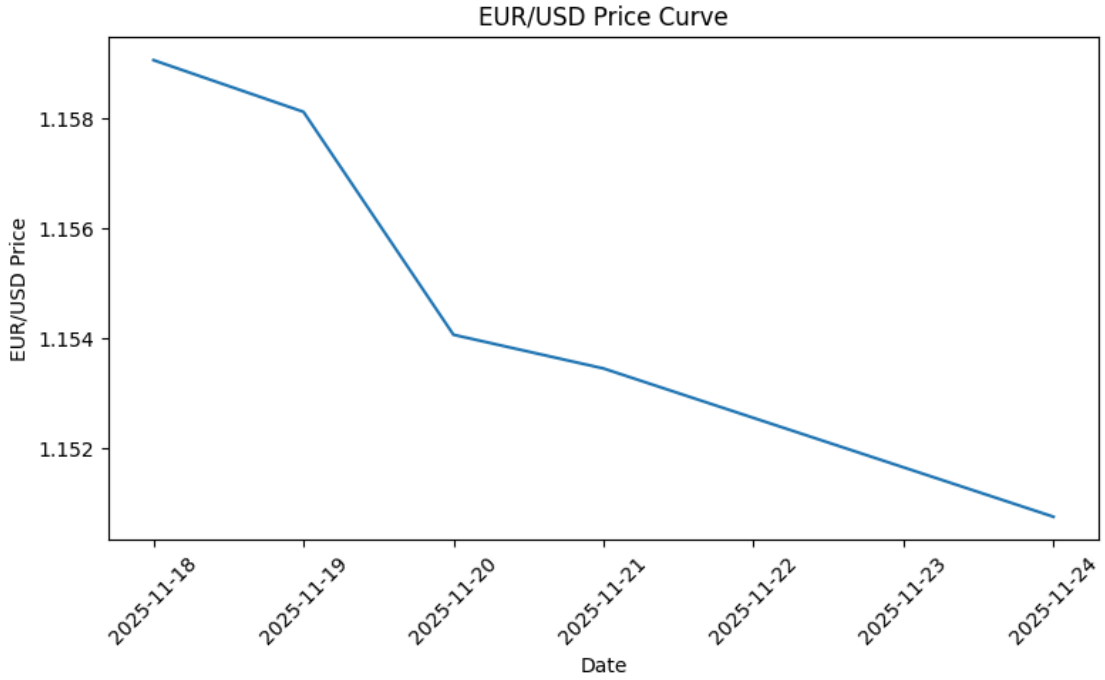
stopping behavior.



The second figure illustrates the daily sentiment curves from Yahoo Finance, ForexLive, and the ECB. Each curve represents the aggregated compound sentiment score for each day from November 18 through November 24. While Yahoo and ECB sentiment remain mostly neutral throughout the period, the ForexLive sentiment curve exhibits more volatility, capturing rapid market commentary. The combined sentiment line is obtained by summing the three sentiment sources, providing an integrated view of market tone.



The third figure presents the EUR/USD price curve corresponding to the same dates. The exchange rate shows a gradual downward trend across the week. This visual correlation between sentiment and exchange rate movement reflects how macroeconomic tone may influence or coincide with currency fluctuations.



Beyond the visualizations, the numerical forecasting results also demonstrate strong performance. Across three separate experimental runs performed on different days, the model produced next-day EUR/USD predictions of 1.1588,1.1522, ,1.1544,1.1568, (Response Date as 11-27,11-24, 11-20,11-19, For all these runs, the difference between predicted and actual values was less than 0.02, indicating that the model captures short-term dynamics effectively and provides consistent, reliable forecasts. The final model produced a next-day prediction of 1.158, closely aligning with observed market data.

After completing this project, we discovered something significant: combining sentiment data from various financial news sources with historical price data can indeed improve the accuracy of short-term currency forecasts.

We cleaned and aggregated the sentiment signals analyzed by VADER, and then fed this sentiment data into an LSTM model. The results showed that the LSTM model not only learned to analyze the price curves in the dataset but also found additional predictive conditions from news sentiment. Therefore, the final prediction results were quite good, with errors generally within 0.02 compared to the actual exchange rate the following day. This demonstrates that teaching AI to use forex news sentiment combined with historical datasets is truly useful for forex forecasting. Interestingly, even using a relatively simple tool like VADER to score sentiment, the improvement in performance is significant when combined with LSTM.

This project also showed us that data preprocessing, feature engineering, and model building are crucial steps in forex forecasting. Of course, there are still many areas for improvement in this project. For example, we could train with data from a longer period, find more diverse news sources, try other different models, and incorporate some key economic indicators. Our project has demonstrated that incorporating sentiment analysis into financial forecasting is entirely feasible, and we have also gained valuable practical experience.

Works cited

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Generative AI Assistance Statement

Since we were unable to save the entire transcript of the conversation with the AI, we could only provide a summary:

Throughout the project, we utilized ChatGPT o4 or other models to assist us in our work. During the coding process, our ideas are all made on our own, but if there were coding difficulties and how to debug, we would let ChatGPT help us, for example, during the preparation of data. During the paper writing process, we ask ChatGPT for data analysis related to the result part, for example, 'How can we integrate more external information?' 'If we want to add more news URLs in our analysis, what should I do for my previous code?'